QBO composite and metrics methodology

July 17, 2026

1 Overview

This document defines a composite framework for characterising the Quasi-Biennial Oscillation (QBO). Westerly and easterly onsets are identified from the detrended, deseasonalised and smoothed 30-hPa zonal wind averaged over 5°S–5°N. Atmospheric fields composited about these onsets are used to diagnose latitudinal width, period, amplitude, phase, vertical extent and descent rate. Individual cycles are compared with the composite using a coherence and correlation metrics.

This methodology is a variation of the QBOi methodology, using 30 hPa as the reference pressure level and extending the compositing method to include latitudinal dependence. See Pahlavan et al. (2021); Simpson et al. (2025); Yu et al. (2017); Lee et al. (2024) for other examples of a QBO composite methodology.

Table 1: Summary of the diagnostic framework.

Diagnostic	Methodological definition
Reference dates	Linearly interpolated zero crossings of the centred five-month running mean of deseasonalised and linearly detrended 30-hPa zonal wind averaged over 5°S–5°N. The default direction is negative to positive (westerly onset); crossings separated by less than five months are averaged.
Composite	Mean deseasonalised anomaly over complete ±15-month cycle windows centred on the reference dates.
Latitudinal width	Full width at half maximum of a fitted parabolic-cylinder structure at each pressure level.
Period, amplitude and phase	Sinusoidal fits to the composite lag profiles at each pressure level. The default QBO period is the mean period from the easterly- and westerly-onset zonal wind composites at 30 hPa.
Vertical extent	Pressure level at 10%-of-maximum of the fitted amplitude profile (with interpolation).
Descent rate	Mean downward displacement of the onset-connected zero-wind contour between successive lags, with a pressure-resolved profile calculated from its crossing lag at each pressure.
Cycle-to-cycle coherence	Squared lag correlation between the raw data and its composite over the default ±15-month cycle window at each pressure level.

Table 2: Key symbols. Detailed definitions in the text.

Symbol	Definition
$u(t, p, \phi)$	Monthly-mean zonal-mean zonal wind.
$u_{30}(t)$	5°S–5°N mean wind at 30 hPa.
$r(t)$	Detrended, deseasonalised and smoothed reference series derived from u_{30}.
x, x'_c	An atmospheric field and its deseasonalised anomaly in QBO cycle c.
C_x	Composite mean of x'_c; for example, C_u is the zonal-wind composite.
$\widehat{C}_x(\phi, p)$	Fitted latitudinal profile of C_x at a specified lag.
$\widehat{C}_x(\tau, p)$	Fitted lag profile of C_x at a specified latitude.

2 Composite methodology

2.1 Reference series and onset dates

Let $u(t, p, \phi)$ be monthly zonal-mean zonal wind, where t is time, p is pressure and ϕ is latitude. The reference series is formed from the 30-hPa wind averaged over 5°S–5°N,

$$u_{30}(t) = \langle u(t, 30\,\mathrm{hPa}, \phi) \rangle_\phi\,, \qquad -5^\circ \leq \phi \leq 5^\circ, \tag{1}$$

where angle brackets denote the latitude average. Wind is interpolated to 30 hPa if that pressure is not available directly. All interpolations are linear in log-pressure coordinates (where applicable).

For monthly analysis, the calendar-month climatology and a linear trend are removed from u_{30} before applying a centred five-month running mean. The result is the reference timeseries $r(t)$, from which reference dates are defined by identifying onset of westerly or easterly winds.

To calculate the QBO period with sub-monthly precision, there is an option to estimate the period using daily means, a cyclic 31-day-smoothed calendar-day climatology and a centred 152-day mean. The 29 February climatology is the mean of those for 28 February and 1 March. A linear trend is removed before smoothing, as for the monthly calculation.

A westerly onset is a negative-to-positive crossing of $r(t)$; an easterly onset is positive to negative. In either case the onset time is interpolated between the two bracketing months where a zero-crossing occurs. If two crossings occur within 5 (nominal) months of each other, the midpoint is used as the reference date. This prevents a short reversal from defining an additional onset. Figure 1 shows the westerly-onset (top) and easterly-onset (bottom) dates and the reference timeseries calculated from ERA5.

2.2 Cycle alignment and compositing

For the composites, we recommend using monthly-mean data. Each interpolated onset date is aligned to its nearest monthly sample and a 31-sample window ($\tau = -15, \ldots, 15$ months) is extracted. QBO cycles without a complete window are omitted. Given a field x, the calendar-month climatology is removed at each pressure and latitude to form the cycle anomaly $x'_c(\tau, p, \phi)$, but the cycle fields are not detrended. Thus detrending affects identification of onset events only. For cycle anomaly x'_c, the composite is

$$C_x(\tau, p, \phi) = \frac{1}{N_c} \sum_{c=1}^{N_c} x'_c(\tau, p, \phi), \tag{2}$$

where N_c is the number of QBO cycles. The onset direction is left implicit in C_x; westerly- and easterly-onset composites are formed separately. Figure 2 shows the westerly-onset zonal wind and

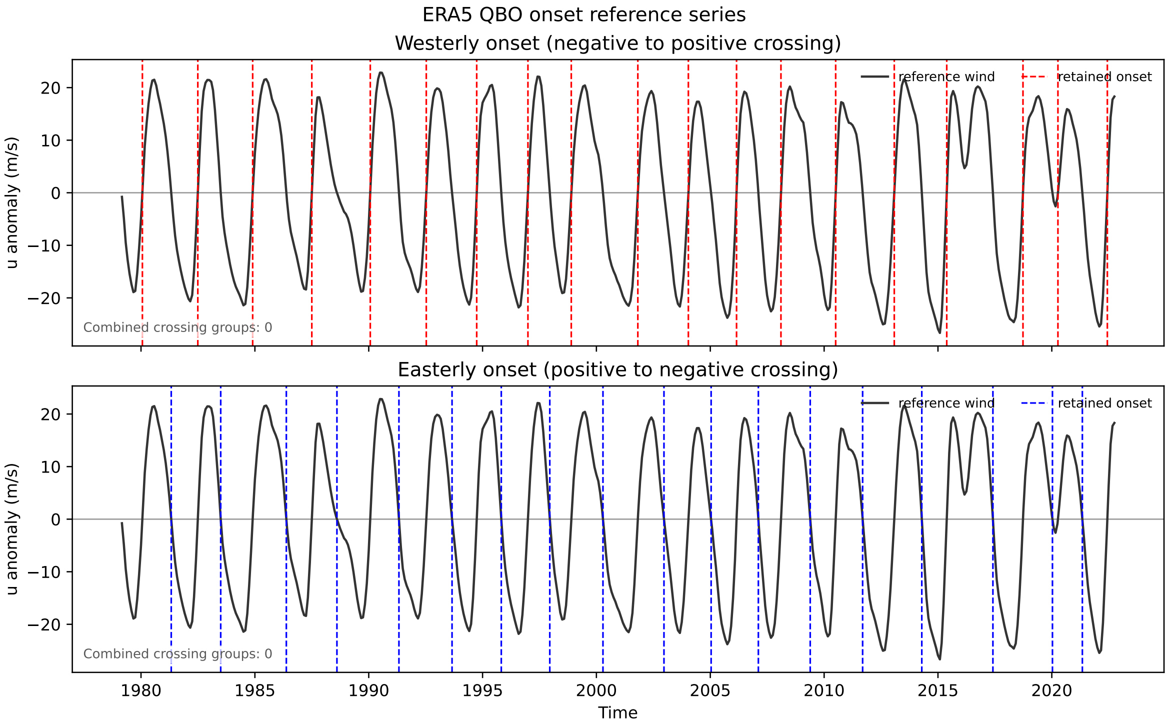

Figure 1: Monthly 30-hPa reference timeseries $r(t)$ for westerly and easterly onsets from ERA5. Dashed lines mark zero crossings. A merged pair of nearby crossings would be shown by an orange interval and dotted original crossings; none occur in this dataset.

temperature composites.

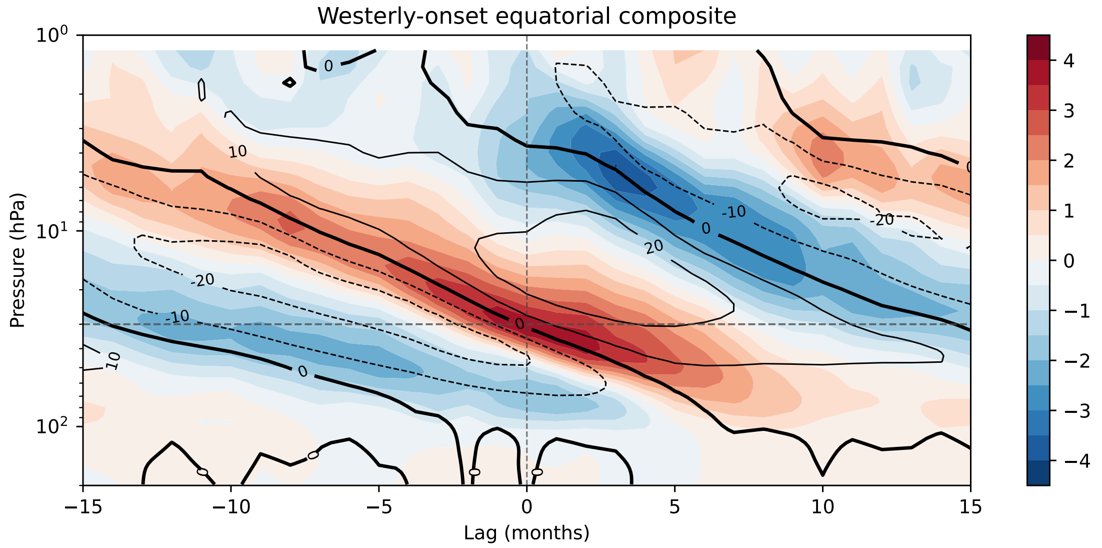

Figure 2: Equatorial westerly-onset composites from ERA5. Filled contours are temperature anomalies (K); line contours are zonal-wind anomalies ($\mathrm{m\,s^{-1}}$). The grey dashed line marks the 30-hPa reference pressure.

3 QBO metrics

3.1 Latitudinal width

Latitudinal structure is diagnosed from the composite profile at zero lag. At each pressure level we fit a sum of parabolic-cylinder functions (Weisstein, n.d.),

$$\widehat{C}_x(\phi, p) = \sum_j a_j(p) D_{n_j}\left(\frac{\phi - c(p)}{s(p)}\right), \tag{3}$$

where $a_j(p)$ is a component amplitude, $c(p)$ and $s(p)$ are the common centre and latitude scale, and j indexes the fitted components. Fits use 30°S–30°N by default, with $|c(p)| \leq 5°$ and $2° \leq s(p) \leq 30°$.

For zonal wind, we use the Gaussian D_0 by default. For temperature and ozone, we use the sum of D_0 and D_2: D_2 looks like a central Gaussian with two side-lobes, with D_0 included to fit the central amplitude. This allows a better fit with both the equatorial and subtropical contributions. Ozone is represented internally as `o3`; CF-compliant input is identified from the standard name `mole_fraction_of_ozone_in_air`.

The latitudinal width is calcualted as the full width at half maximum (FWHM) of the absolute fitted equatorial lobe. For completeness, we also report the scale $s(p)$ (which, if equivalent to the FWHM, would be half its value). A fit is accepted when its RMS residual, normalised by the full-cycle lag–latitude RMS after removal of the lag mean, is at most 0.5 and its log-FWHM differs by at most 0.1 from interpolation between adjacent presssure levels. This is designed to reject poor fits between the QBO jets and in the upper troposphere/lower stratosphere where the QBO amplitude weakens. Both thresholds are configurable. Figure 3 shows the latitudinal width calculation for zonal wind at 50 hPa and temperature at 30 hPa from ERA5.

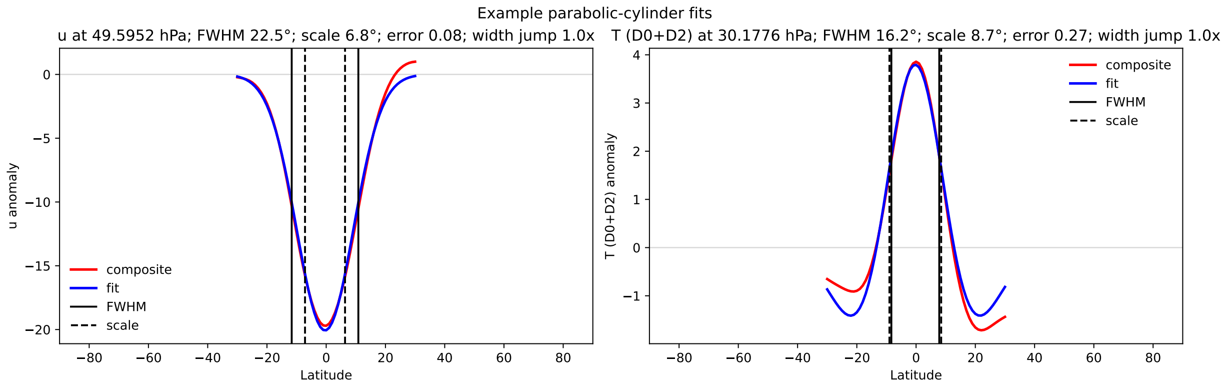

Figure 3: Parabolic-cylinder fits to zonal wind near 50 hPa and temperature near 30 hPa from ERA5. Solid and dashed markers show FWHM and fitted scale, respectively.

3.2 Period, amplitude, phase, and vertical extent

At each pressure level, the composite lag profile is fitted with

$$\widehat{C}_x(\tau, p) = b(p) + A(p)\sin\left[\frac{2\pi\tau}{P} + \theta(p)\right], \tag{4}$$

where $b(p)$ is the offset, $A(p) > 0$ is the amplitude, P is the QBO period and $\theta(p) \in [-\pi, \pi)$ is the phase.

The QBO period is the mean period obtained from fits to the easterly- and westerly-onset zonal wind composites C_u at 30 hPa, interpolated in pressure if necessary. This period is held fixed at all pressure levels for subsequent calculations of the amplitude and phase for a given variable. The resulting amplitude and phase profiles therefore describe one common oscillation frequency. The QBO period is continuous rather than rounded to the sampling period of the underlying composite. If only monthly raw data is supplied, the sub-monthly precision of the period is therefore inferred.

If daily data is supplied, the QBO period can optionally be calculated by applying the same compositing and fitting procedure to daily raw data. Nominal-month parameters are converted using 365.25/12 days per month: the reference smoothing is 152 days and the $\pm$15-month cycle window becomes $\pm$457 days. Each interpolated onset time is aligned to its nearest daily sample. The unsmoothed, deseasonalised 30-hPa wind at the equator is then composited and fitted as a function of lag in days. As for monthly cycle fields, these daily anomalies are not detrended, and the final reported period is an average of the calculation for a westerly- and easterly-onset composite. Figure 4 shows the fit calculation on daily-sampled composites.

The vertical extent of the QBO is defined as the linearly interpolated pressure level where the amplitude $A(p)$ is 10% of its maximum:

$$A(p_{\text{top}}) = 0.1 \max_{p}\{A(p)\}. \tag{5}$$

No value is assigned if it the threshold is not met in the range of pressure levels provided. Figure 5 shows the phase and amplitude profiles for zonal wind and temperature using the daily-sampled QBO period of 27.96 months with the vertical extent and the reference height (30 hPa) indicated.

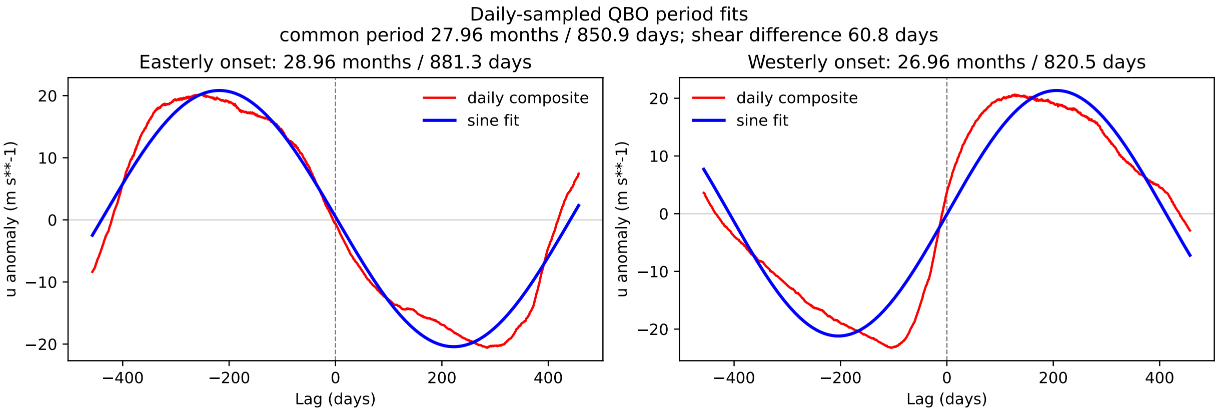

Figure 4: Daily ERA5 30-hPa composites and sine fits. The periods for different choices of onset differ by 60.8 days and average to 850.9 days (27.96 months).

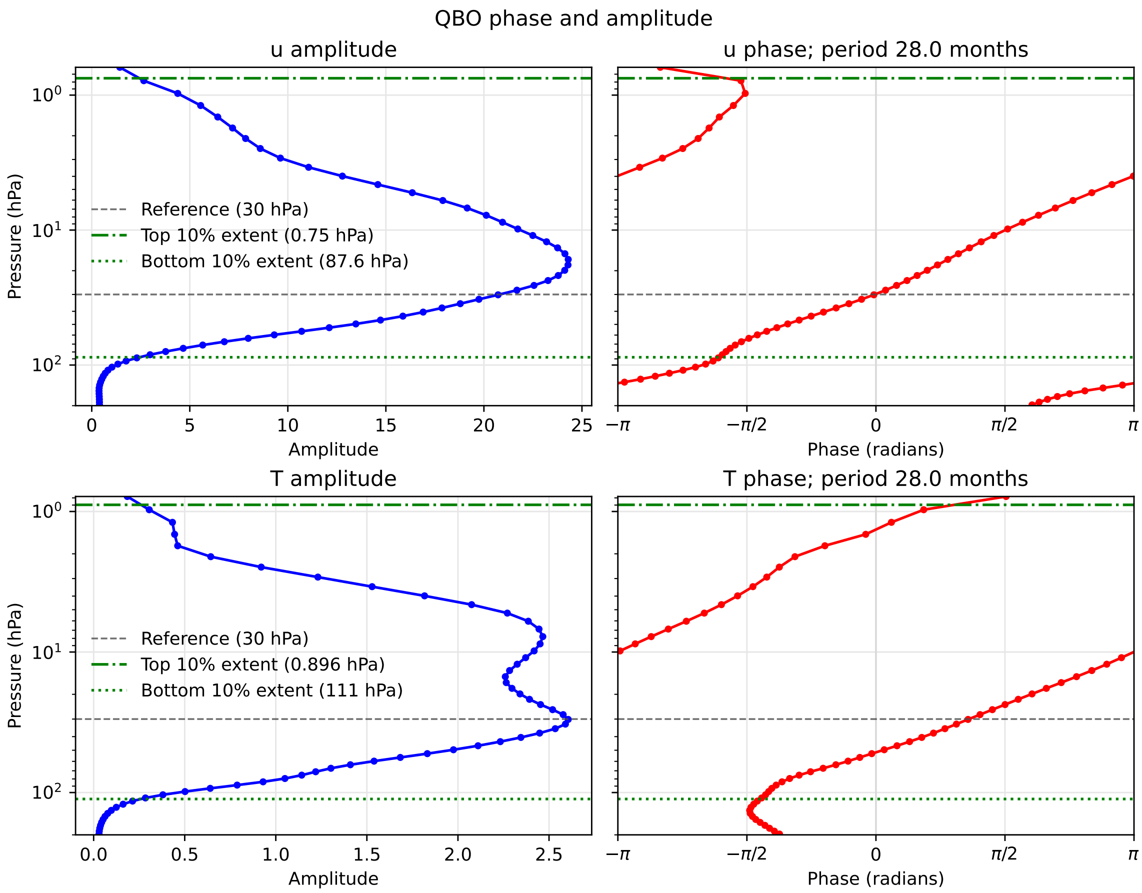

Figure 5: Amplitude and phase profiles for a QBO period of 27.96 months from ERA5. Grey dashed and green dash–dot lines mark 30 hPa and the vertical extent, respectively.

3.3 Descent rate

The mean descent rate follows Schenzinger et al. (2017), adapted to pressure coordinates and the zonal-wind composite C_u. For each onset direction, the zero-wind contour passing through the 30-hPa onset is followed by linear interpolation. Its pressure $p_0(\tau)$ gives the downward displacement between successive monthly lags,

$$v_{d,i} = \frac{p_0(\tau_{i+1}) - p_0(\tau_i)}{\tau_{i+1} - \tau_i}, \tag{6}$$

where positive $v_{d,i}$ indicates downward motion. The reported mean is the arithmetic mean of its positive monthly values.

A pressure-resolved rate is obtained from the lag $\tau_0(p)$ at which the same contour crosses each pressure level. At interior levels a centred difference gives

$$v_d(p_j) = \frac{p_{j+1} - p_{j-1}}{\tau_0(p_{j+1}) - \tau_0(p_{j-1})}, \tag{7}$$

with one-sided differences at the ends. Only the monotonically descending part of the contour is retained, with no extrapolation beyond its diagnosed pressure range. The calculation is applied separately to easterly- and westerly-onset composites between 1 and 200 hPa by default. The positive monthly rates from each individual QBO cycle are interpolated onto the common pressure levels to calculate the cycle-to-cycle standard deviation.

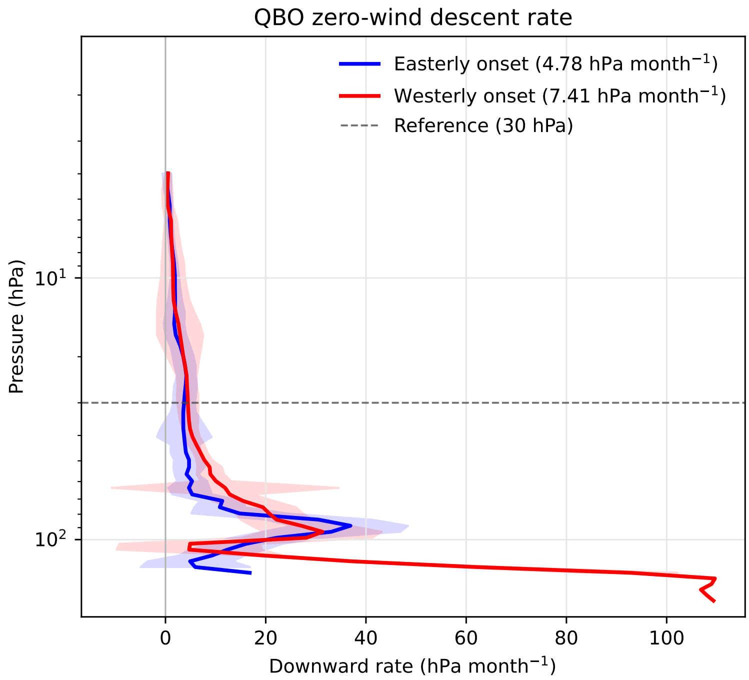

Figure 6: Pressure-resolved downward rates of the onset-connected zero-wind contours in the ERA5 easterly- and westerly-onset composites. Legend values are the mean monthly displacements. Shading spans one standard deviation across individual QBO cycles and the grey dashed line marks 30 hPa.

3.4 Cycle-to-cycle coherence

For each equatorial zonal-wind cycle u'_c (with a ±15-month window by default) and pressure level, coherence is the squared Pearson correlation with C_u across lag,

$$\mathcal{C}_{c,p} = \mathrm{corr}_\tau \left[u'_c(\tau, p), C_u(\tau, p)\right]^2. \tag{8}$$

The coherence is bounded by $0 \leq \mathcal{C}_{c,p} \leq 1$ and measures similarity of temporal evolution between individual cycles and the QBO composite. The cycle mean is weighted by pressure-layer thickness in log pressure,

$$w_p = \frac{|\nabla \log p|}{\sum_p |\nabla \log p|}. \tag{9}$$

The vertically averaged coherence of cycle c is therefore

$$\overline{\mathcal{C}}_c = \sum_p w_p \mathcal{C}_{c,p}. \tag{10}$$

Figure 7 shows the QBO wind anomalies and cycle/composite coherence.

Alongside coherence, we use two metrics to capture the cycle-to-cycle variability in QBO amplitude, structure and timing. For any field $q(\tau, p)$, define the weighted mean

$$\langle q \rangle_W = \frac{1}{N_\tau} \sum_\tau \sum_p w_p q(\tau, p), \tag{11}$$

and let $\mu_c = \langle u'_c \rangle_W$ and $\mu_{C_u} = \langle C_u \rangle_W$. The whole-pattern correlation is

$$R_c = \frac{\langle (u'_c - \mu_c)(C_u - \mu_{C_u}) \rangle_W}{\sqrt{\langle (u'_c - \mu_c)^2 \rangle_W \, \langle (C_u - \mu_{C_u})^2 \rangle_W}}. \tag{12}$$

It is a signed measure on $[-1, 1]$: positive values indicate similar lag–pressure structure and timing, while negative values indicate an opposite pattern.

The RMS-amplitude ratio is

$$A_c^{\mathrm{RMS}} = \frac{\sqrt{\langle (u'_c)^2 \rangle_W}}{\sqrt{\langle C_u^2 \rangle_W}}. \tag{13}$$

A value of one gives equal RMS amplitude; values above and below one indicate stronger and weaker cycles, respectively. Both metrics are dimensionless. Figure 8 plots R_c against A_c^{RMS} for each cycle, with onset year shown by colour and the point label. At each lag and pressure, values outside the standard 1.5-IQR limits are excluded from the cycle standard deviation; the composite itself retains all cycles.

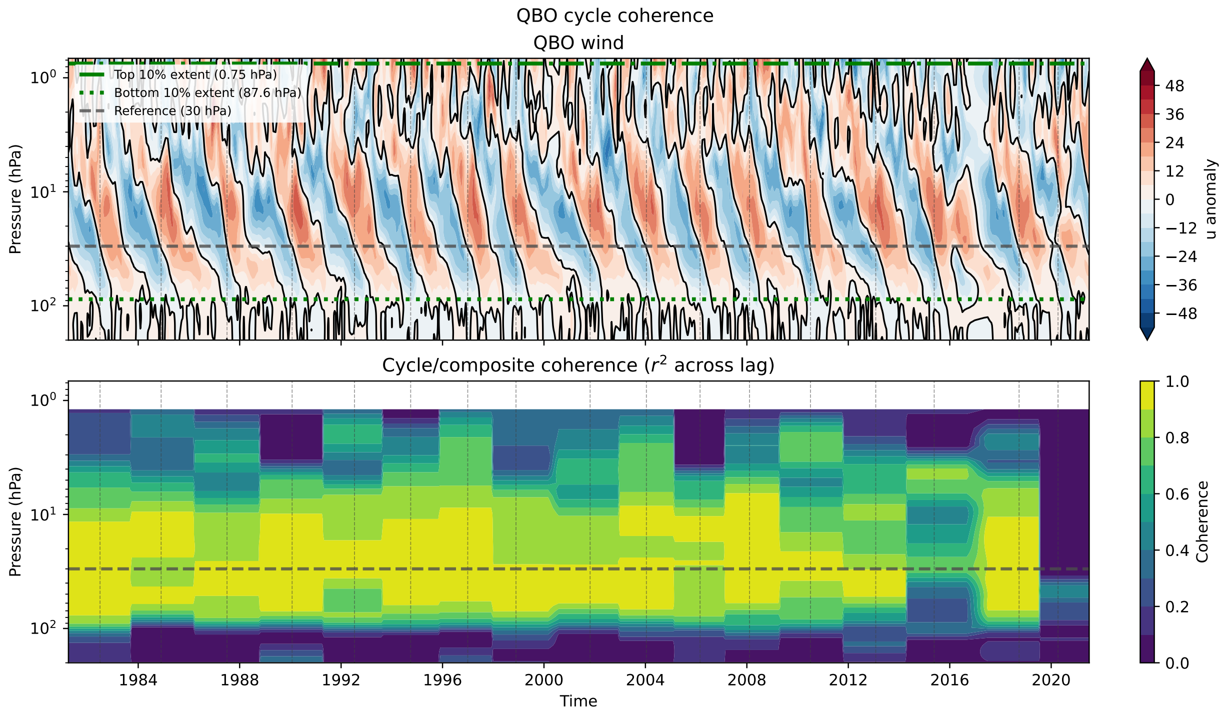

Figure 7: QBO-cycle anomalies and pressure-resolved coherence on a continuous timeline. Vertical dashed lines mark westerly onsets; the horizontal grey dashed line marks 30 hPa. Green dash–dot and dotted lines in the upper panel mark the top and bottom vertical extent, respectively.

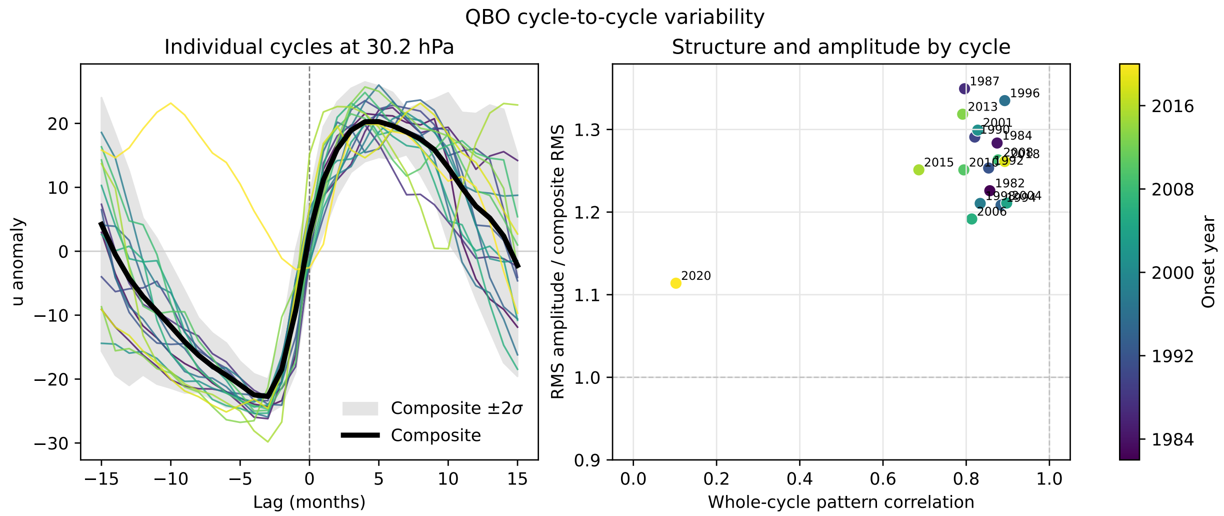

Figure 8: Westerly-onset QBO cycles at 30 hPa and their amplitude, timing and structure metrics from ERA5. The left panel shows the individual onset-aligned cycles (coloured by onset year), their composite (black), and the composite plus or minus two standard deviations across cycles (grey shading). The right panel shows whole-pattern correlation against RMS-amplitude ratio; colour and labels give onset year. Dashed lines mark zero lag in the left panel and unit correlation and amplitude ratio in the right panel. The standard deviation excludes pointwise values outside the standard 1.5-IQR limits.

4 Configurable parameters

Table 3: User-configurable methodology parameters and defaults.

Parameter	Default	Role
Reference timeseries		
Reference pressure	30 hPa	Onset-defining level; interpolated if absent.
Reference latitude band	5°S–5°N	Band averaged for the reference timeseries.
Reference smoothing	5 months	Centred running-mean for monthly data. Daily data uses an equivalent 152-sample running mean.
Daily climatology smoothing	31 days	Cyclic mean of the calendar-day climatology.
Crossing direction	Westerly	Negative-to-positive; easterly is the reverse.
Crossing merge tolerance	5 months	Minimum separation between onset events.
Composites		
Cycle half-window	15 months	31 monthly lags or ±457 daily lags.
Composite deseasonalisation	On	Remove the calendar-month climatology.
Period		
Daily period latitude	0°	Latitude to fit for the daily period estimate.
Initial period guess	28 months	Initial value; the fitted period is bounded between half and twice this value.
Latitudinal width		
Parabolic-cylinder orders	D_0 (u); $D_0 + D_2$ (T, O_3)	Variable-dependent default.
Fit latitude range	30°S–30°N	Calculation range.
Fit pressure range	1–200 hPa	Calculation range.
Maximum normalised fit error	0.5	Residual-to-full-cycle-RMS threshold; 0.3 is used for wind in the ERA5 illustration.
Maximum log-width discontinuity	0.1	Vertical-continuity threshold for log-FWHM.
Phase & amplitude		
Supplied QBO period	None	If absent, estimate from the 30-hPa composites; the daily period estimate may instead be supplied.
Period reference pressure	30 hPa	Height of the QBO period fit.
Phase/amplitude pressure range	1–200 hPa	Calculation range.
Vertical extent		
Vertical-extent threshold	0.10	Fraction of maximum amplitude at the upper boundary.
Descent rate		
Descent pressure range	1–200 hPa	Calculation range.
Descent reference pressure	30 hPa	Selects the onset-connected zero-wind contour.
Coherence		
Coherence pressure range	1–200 hPa	Calculation range.

5 Illustration with ERA5

The figures use ERA5 for 1979–2022: daily wind for the optional period estimate and monthly data otherwise. Of 19 westerly and 18 easterly onsets, 17 and 18, respectively, have complete cycle windows. Table 4 illustrates the resulting metrics; the values are not prescribed observational targets.

Table 4: Results from ERA5 data.

Quantity	ERA5 illustration
Complete westerly/easterly cycle windows	17 / 18
Monthly-sampled QBO period (easterly-/westerly-onset)	28.96 / 26.96 months
Daily-sampled QBO period (average of both onsets)	27.96 months / 850.9 days
Easterly–westerly period difference	60.8 days
Vertical extent, easterly/westerly composites	0.82 / 0.75 hPa
Mean descent rate, easterly-/westerly-onset composite	4.78 / 7.41 $\text{hPa}\,\text{month}^{-1}$
Mean westerly-onset cycle coherence	0.57
Westerly-onset wind FWHM near 50 hPa	22.5°
Westerly-onset temperature FWHM near 30 hPa	16.2°

References

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